

Chapter 8: Simple Linear Regression

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Regression Analysis

Problem: In scientific, technical, medical, economic, and social activities, there is a need to determine the relationship between two or more random variables.

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To address these issues, we use the **Regression Analysis** technique.

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 - ▶ The relationship between $X_1, \dots, X_p$ and Y is represented by a linear function, i.e.,

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- ▶ Based on determining the relationship between the dependent variable Y and explanatory variables $X_1, X_2, \dots, X_p$, we can:
 - ▶ predict the value of Y ,
 - ▶ explain the impact of changes in explanatory variables on the dependent variable.

Simple linear regression model

Definition 1.1

A *simple linear regression model* related to a random variable Y and an explanatory variable x is an equation of the form

$$Y = \beta_0 + \beta_1 X + \epsilon, \tag{1}$$

where

- β_0, β_1 are unknown parameters, called regression coefficients,
- X is the independent variable, explaining y ,
- ϵ is the error term.

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$$\mathbb{E}(Y|X = x) = \beta_0 + \beta_1 x. \quad (2)$$

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Hence, the conditional distribution of Y given $X = x$ is

$$Y|X = x \sim \mathcal{N}(\beta_0 + \beta_1 x, \sigma^2) \quad (3)$$

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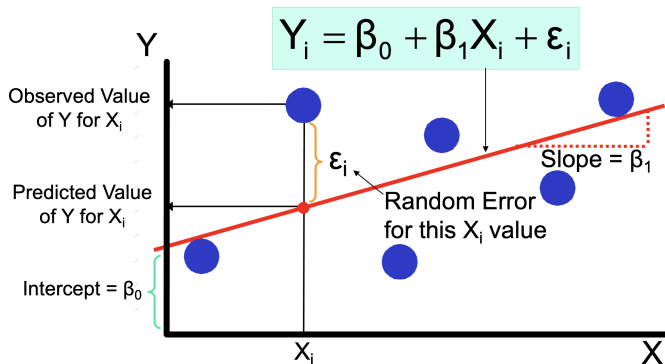
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 - ▶ The impact of other factors (not X): the error term ϵ .
- ▶ With $(x_1, y_1), \dots, (x_n, y_n)$ being n pairs of observed values of a random sample of size n , from (5) we have

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, 2, \dots, n \quad (4)$$

Simple linear regression model

- Use a **scatter plot** to represent the pairs of observed values (x_i, y_i) on the coordinate system Oxy .



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- ▶ A good-fitting line should be "close to the data points".
- ▶ Finding $\hat{\beta}_0$ and $\hat{\beta}_1$: use the **Method of Least Squares**.

Method of least squares

- For data $(x_i, y_i), i = 1, \dots, n$, from (5) we have

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i. \quad (6)$$

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- The difference between the observed value y_i and the predicted value $\hat{y}_i$ is called the residual of the i th observation, defined as follows

$$e_i = y_i - \hat{y}_i = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i). \quad (7)$$

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Definition 2.1

Sum of Squares for Errors (SSE) or sum of squares of residuals for n data points is defined as follows

$$\text{SSE} = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n \left[y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i) \right]^2. \quad (8)$$

The essence of the method of least squares is to find the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ so that SSE attains the minimum value.

Method of least squares

From (8), taking derivatives with respect to β_0 and β_1 ,

$$\frac{\partial \text{SSE}}{\partial \beta_0} = -2 \sum_{i=1}^n [y_i - (\beta_0 + \beta_1 x_i)] = 0,$$

$$\frac{\partial \text{SSE}}{\partial \beta_1} = -2 \sum_{i=1}^n [y_i - (\beta_0 + \beta_1 x_i)] x_i = 0,$$

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we obtain the system of equations

$$\begin{aligned} n\beta_0 + \beta_1 \sum_{i=1}^n x_i &= \sum_{i=1}^n y_i, \\ \beta_0 \sum_{i=1}^n x_i + \beta_1 \sum_{i=1}^n x_i^2 &= \sum_{i=1}^n x_i y_i. \end{aligned} \tag{9}$$

Least squares estimators

Solving the system (9), we find the least squares estimators of β_0 and β_1 as

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - \frac{(\sum_{i=1}^n x_i)(\sum_{i=1}^n y_i)}{n}}{\sum_{i=1}^n x_i^2 - \frac{(\sum_{i=1}^n x_i)^2}{n}} = \frac{S_{xy}}{S_{xx}}, \quad (10)$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}. \quad (11)$$

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with S_{xx} and S_{xy} defined as

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - \frac{(\sum_{i=1}^n x_i)^2}{n}, \quad (12)$$

$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - \frac{(\sum_{i=1}^n x_i)(\sum_{i=1}^n y_i)}{n}. \quad (13)$$

Least squares estimators

- ▶ $\hat{\beta}_0$ and $\hat{\beta}_1$ are called the least squares estimators.
- ▶ The line $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ is called the least squares regression line, satisfying the following properties:

(1)

$$\text{SSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2,$$

attains the minimum value,

(2)

$$\text{SE} = \sum_{i=1}^n (y_i - \hat{y}_i) = \sum_{i=1}^n e_i = 0,$$

where SE is the sum of errors.

Properties of least squares estimators

Theorem 3.1

Consider $Y = \beta_0 + \beta_1 X + \epsilon$ as a simple linear regression model with $\epsilon \sim \mathcal{N}(0, \sigma^2)$. For n pairs of observed values $(x_i, y_i), i = 1, \dots, n$ we have

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i.$$

Let $\hat{\beta}_0$ and $\hat{\beta}_1$ be the estimates of β_0 and β_1 obtained from the least squares method, then

1. $\hat{\beta}_0$ and $\hat{\beta}_1$ follow normal distribution.
2. The expectations and variances of $\hat{\beta}_0$ and $\hat{\beta}_1$ are given by

$$\mathbb{E}(\hat{\beta}_0) = \beta_0, \text{Var}(\hat{\beta}_0) = \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) \sigma^2, \quad (14)$$

$$\mathbb{E}(\hat{\beta}_1) = \beta_1, \text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}. \quad (15)$$

The estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased estimates for β_0 and β_1 .

Properties of least squares estimators

Proposition 3.2

The estimate of the variance σ^2 of the model errors is given by

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n - 2}. \quad (16)$$

Also, we have $\mathbb{E}[\hat{\sigma}^2] = \sigma^2$, meaning that $\hat{\sigma}^2$ is an unbiased estimator for σ^2 .

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Definition 3.3

*In a simple linear regression model, the **standard errors (SE)** of the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are defined as*

$$\text{SE}(\hat{\beta}_0) = \sqrt{\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right) \hat{\sigma}^2}, \quad (17)$$

$$\text{SE}(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}, \quad (18)$$

Properties of least squares estimators

Theorem 3.4 (Gauss - Markov)

Consider a simple linear regression model

$$Y = \beta_0 + \beta_1 X + \epsilon,$$

with $\hat{\beta}_0$ and $\hat{\beta}_1$ being the least squares estimators for β_0 and β_1 , then $\hat{\beta}_0$ and $\hat{\beta}_1$ are the best unbiased estimators.

Measures of variation

Given:

- ▶ SST: Total Sum of Squares,

$$\text{SST} = \sum_{i=1}^n (y_i - \bar{y})^2.$$

Also denoted as S_{yy} .

- ▶ SSR: Regression Sum of Squares,

$$\text{SSR} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2.$$

- ▶ SSE: Error Sum of Squares,

$$\text{SSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

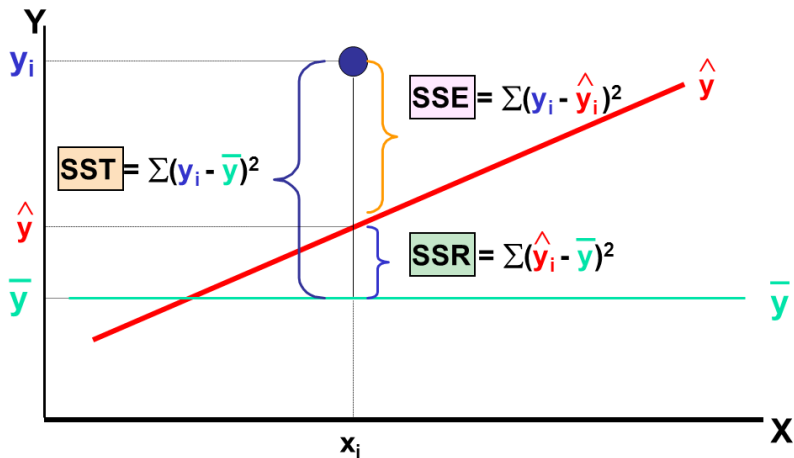
Measures of variation

- ▶ SST: measures the variability of the values y_i around the center value of the data $\bar{y}$,
- ▶ SSR: explains the variability related to the linear relationship between X and Y ,
- ▶ SSE: explains the variability due to other factors (not related to the linear relationship between X and Y).

We can show that:

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad (19)$$
$$\text{SST} = \text{SSR} + \text{SSE}.$$

Measures of variation



Coefficient of determination

Definition 4.1

The *Coefficient of determination* is the ratio of the total sum of squares caused by the variation of the dependent variable due to the variation of the independent variables (explanatory variables) to the total overall sum of squares.

The coefficient of determination is often referred to as *R* - squared (R^2). Its formula is given by:

$$R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{\text{SSR}}{\text{SSR} + \text{SSE}}. \quad (20)$$

Note: $0 \leq R^2 \leq 1$.

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The coefficient of determination of a regression model allows us to evaluate whether the model obtained explains the relationship between the dependent variable Y and the independent variable X well.

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► Calculation of SSR:

$$\begin{aligned}\text{SSR} &= \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 x_i - \bar{y})^2 = \sum_{i=1}^n (\bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_i - \bar{y})^2 \\ &= \sum_{i=1}^n (\hat{\beta}_1 x_i - \hat{\beta}_1 \bar{x})^2 = \hat{\beta}_1^2 \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= \hat{\beta}_1^2 S_{xx} = \hat{\beta}_1 \hat{\beta}_1 S_{xx} = \hat{\beta}_1 \frac{S_{xy}}{S_{xx}} S_{xx} \\ &= \hat{\beta}_1 S_{xy}.\end{aligned}$$

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$$R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{\text{SSR}}{\text{SSR} + \text{SSE}}$$

► Calculation of SSR:

$$\begin{aligned}\text{SSR} &= \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 x_i - \bar{y})^2 = \sum_{i=1}^n (\bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_i - \bar{y})^2 \\ &= \sum_{i=1}^n (\hat{\beta}_1 x_i - \hat{\beta}_1 \bar{x})^2 = \hat{\beta}_1^2 \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= \hat{\beta}_1^2 S_{xx} = \hat{\beta}_1 \hat{\beta}_1 S_{xx} = \hat{\beta}_1 \frac{S_{xy}}{S_{xx}} S_{xx} \\ &= \hat{\beta}_1 S_{xy}.\end{aligned}$$

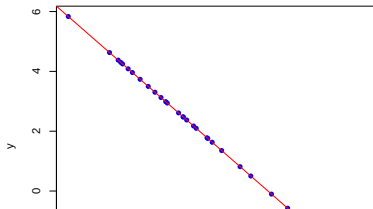
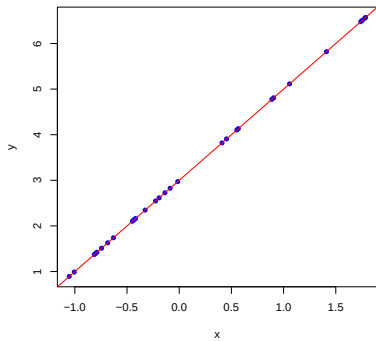
► Calculation of SSE:

$$\text{SSE} = \text{SST} - \text{SSR} = \text{SST} - \hat{\beta}_1 S_{xy},$$

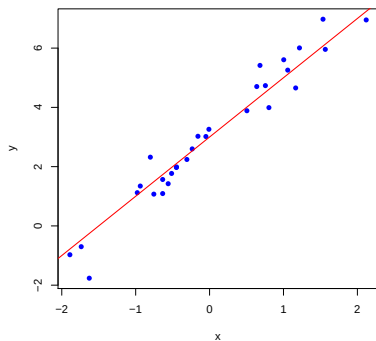
with

$$\text{SST} = S_{yy} = \sum_{i=1}^n y_i^2 - \frac{(\sum_{i=1}^n y_i)^2}{n}.$$

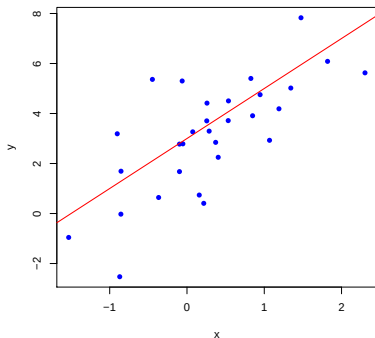
R^2 and the relationship between X and Y



R^2 and the relationship between X and Y

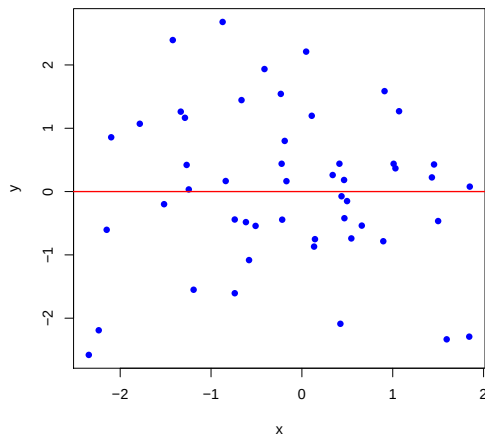


- ▶ As R^2 gets closer to 1, the linear relationship between X and Y becomes stronger. The majority of the variation in Y is explained by X .



- ▶ As R^2 gets closer to 0, the linear relationship between X and Y becomes weaker. The variation in Y is less explained by X .

R^2 and the relationship between X and Y



- $R^2 = 0$: there is no linear relationship between X and Y . No variation in Y is explained by X .

Linear Regression: example

Example 5.1

A botanist examines the relationship between the total leaf area (in cm^2) of soybean plants and their dry weight (in gram). The botanist plants 13 soybean plants in a greenhouse and measures the total leaf area and weight of these plants after 16 days, with the results given in the table:

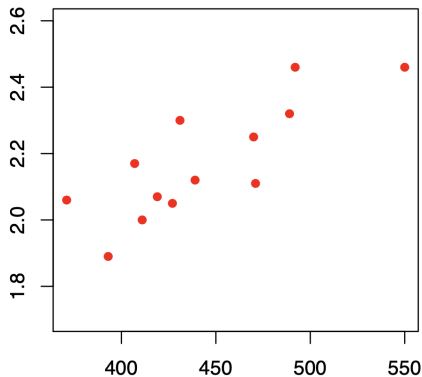
x_i	411	550	471	393	427	431	492	371	470	419	407	48
y_i	2.00	2.46	2.11	1.89	2.05	2.30	2.46	2.06	2.25	2.07	2.17	2.3

- (a) Plot the scatter diagram representing the leaf area X and the dry weight Y of soybean plants with the given sample.
- (b) Find the regression line representing the relationship between the weight Y and the leaf area X . Plot the regression line on the scatter plot.
- (c) Calculate the R^2 coefficient and comment on the model.

Linear Regression: example

Solution to example 5.1:

(a) Plot the scatter diagram:



Linear Regression: example

Solution to example 5.1:

(b) Find the estimated regression line

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

Recall that the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are calculated by:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}},$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

with

$$S_{xx} = \sum_{i=1}^n x_i^2 - \frac{(\sum_{i=1}^n x_i)^2}{n},$$

$$S_{xy} = \sum_{i=1}^n x_i y_i - \frac{(\sum_{i=1}^n x_i)(\sum_{i=1}^n y_i)}{n}.$$

Linear Regression: example

Solution to example 5.1:

(b) From the table, we calculate

$$\begin{aligned}\sum_{i=1}^n x_i &= 5770, & \sum_{i=1}^n x_i^2 &= 2589458, \\ \sum_{i=1}^n y_i &= 28.26, & \sum_{i=1}^n x_i y_i &= 12625.99.\end{aligned}$$

Linear Regression: example

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Thus,

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = 443.8462, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i = 2.1738,$$

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$$S_{xx} = 28465.69, \quad S_{xy} = 82.8977.$$

Linear Regression: example

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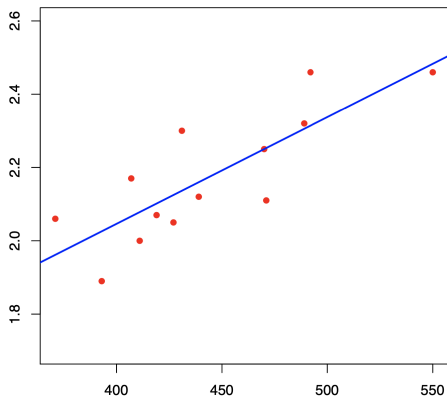
Therefore,

$$\begin{aligned}\hat{\beta}_1 &= \frac{S_{xy}}{S_{xx}} = \frac{82.8977}{28465.69} = 0.002912, \\ \hat{\beta}_0 &= \bar{y} - \hat{\beta}_0 \bar{x} = 2.1738 - 0.002912 \times 443.8462 = 0.8813.\end{aligned}$$

Linear Regression: example

Solution to example 5.1:

(b) Plot the fitted line on the scatter plot:



Linear Regression: example

Solution to example 5.1:

- (c) Calculate the coefficient of determination R^2 : recall the formula for the coefficient of determination

$$R^2 = \frac{\text{SSR}}{\text{SST}},$$

Linear Regression: example

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$$R^2 = \frac{SSR}{SST},$$

where

$$SST = S_{yy} = \sum_{i=1}^n y_i^2 - \frac{(\sum_{i=1}^n y_i)^2}{n} = 0.3637,$$

and

$$SSR = \hat{\beta}_1 S_{xy} = 0.002912 \times 82.8977 = 0.2414.$$

Therefore,

$$R^2 = \frac{SSR}{SST} = \frac{0.2414}{0.3637} = 0.6637.$$

Standard Errors of $\hat{\beta}_0$ and $\hat{\beta}_1$

- Recall that we have the standard errors (SE) of $\hat{\beta}_0$ and $\hat{\beta}_1$ given by

$$\text{SE}(\hat{\beta}_0) = \sqrt{\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right) \hat{\sigma}^2},$$

$$\text{SE}(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}},$$

where

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n - 2}.$$

Confidence intervals for β_0 and β_1

100(1 - α)% confidence intervals for the regression coefficients β_1 and β_0 are given by:

- ▶ 100(1 - α)% confidence interval for β_1 :

$$\hat{\beta}_1 - t_{\alpha/2}^{n-2} \text{SE}(\hat{\beta}_1) \leq \beta_1 \leq \hat{\beta}_1 + t_{\alpha/2}^{n-2} \text{SE}(\hat{\beta}_1) \quad (21)$$

- ▶ 100(1 - α)% confidence interval for β_0 :

$$\hat{\beta}_0 - t_{\alpha/2}^{n-2} \text{SE}(\hat{\beta}_0) \leq \beta_0 \leq \hat{\beta}_0 + t_{\alpha/2}^{n-2} \text{SE}(\hat{\beta}_0) \quad (22)$$

Hypothesis testing for regression coefficients

Problem:

- Suppose we need to build a regression model with the dependent variable Y and a set of explanatory variables $X_1, X_2, \dots, X_p$.

Hypothesis testing for regression coefficients

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- ▶ We can consider a general multiple linear regression model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \epsilon.$$

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- ▶ To determine which variables are significant in the model, we can perform hypothesis tests for the corresponding regression coefficients. Specifically,

$$H_0 : \beta_j = 0 \quad \text{vs} \quad H_1 : \beta_j \neq 0,$$

for $j = 0, \dots, p$.

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- ▶ In the current course content, we are examining a simple linear regression model $Y = \beta_0 + \beta_1 X + \epsilon$, so we will consider hypothesis testing for β_0 and β_1 .

Hypothesis testing for β_0

- The hypothesis testing problem for the intercept β_0 in a simple linear regression model is formulated as follows:

$$\begin{cases} H_0 : \beta_0 = b_0 \\ H_1 : \beta_0 \neq b_0 \end{cases}$$

with a specified value b_0 and significance level α . Typically $b_0 = 0$.

Hypothesis testing for β_0

Testing procedure

1. State the null hypothesis H_0 and alternative hypothesis H_1 ,

Hypothesis testing for β_0

Testing procedure

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2. Determine the significance level α ,

Hypothesis testing for β_0

Testing procedure

1. State the null hypothesis H_0 and alternative hypothesis H_1 ,
2. Determine the significance level α ,
3. Calculate the test statistic:

$$t_{\beta_0} = \frac{\hat{\beta}_0 - b_0}{\text{SE}(\hat{\beta}_0)}, \quad \text{with } \text{SE}(\hat{\beta}_0) = \sqrt{\hat{\sigma}^2 \left(1 + \frac{\bar{x}^2}{S_{xx}} \right)}.$$

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4. Reject H_0 if: $|t_{\beta_0}| > t_{1-\alpha/2}^{n-2}$.

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5. Conclusion: Reject H_0 / Fail to reject H_0 .

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5. Conclusion: Reject H_0 / Fail to reject H_0 .
6. Alternatively, we can use the p -value defined by

$$p = 2\mathbb{P}(T_{n-2} \geq |t_{\beta_0}|),$$

and reject H_0 when $p \leq \alpha$.

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Sample correlation coefficient

Definition 8.1

Consider two random variables X, Y . The *covariance* of X and Y , denoted by $\text{Cov}(X, Y)$, is defined as

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))] = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y). \quad (23)$$

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Definition 8.2

The *correlation coefficient* of two random variables X and Y , denoted by ρ_{XY} , is defined as

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}. \quad (24)$$

For any two random variables X and Y : $-1 \leq \rho_{XY} \leq 1$.

Sample correlation coefficient

Definition 8.3

For a sample of size n : $(x_i, y_i), i = 1, \dots, n$, the sample correlation coefficient, denoted by r_{XY} , is defined as

$$r_{XY} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}. \quad (25)$$

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- ▶ $0 < r_{XY} \leq 1$: positive correlation. The closer r_{XY} is to 1 , the stronger the positive linear relationship between X and Y .

Sample correlation coefficient

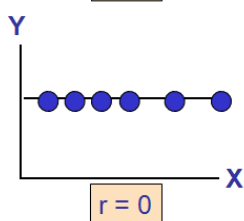
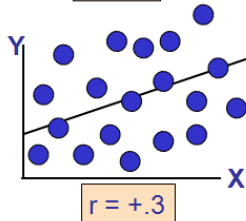
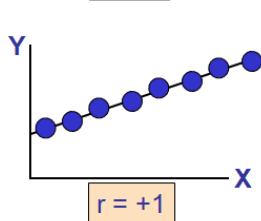
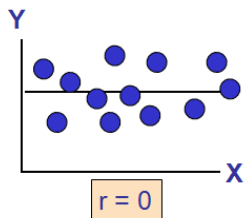
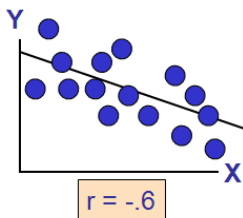
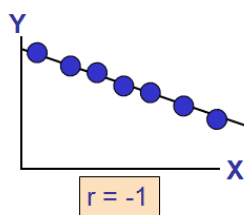
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- ▶ $0 < r_{XY} \leq 1$: positive correlation. The closer r_{XY} is to 1 , the stronger the positive linear relationship between X and Y .
- ▶ As r_{XY} approaches 0 , it indicates a weak linear relationship. $r_{XY} = 0$: no linear relationship between X and Y .

Evaluating the sample correlation coefficient



Relationship between sample correlation coefficient and coefficient of determination

- Note that,

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \sqrt{\frac{S_{yy}}{S_{xx}}} \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \sqrt{\frac{S_{yy}}{S_{xx}}} r_{XY} = \sqrt{\frac{SST}{S_{xx}}} r_{XY},$$

since S_{yy} is exactly SST.

- Hence,

$$r_{XY}^2 = \hat{\beta}_1^2 \frac{S_{xx}}{SST} = \hat{\beta}_1 \frac{S_{xy}}{S_{xx}} \frac{S_{xx}}{SST} = \frac{\hat{\beta}_1 S_{xy}}{SST} = \frac{SSR}{SST} = R^2.$$

- Therefore, the coefficient of determination R^2 of the simple linear regression model is equal to the square of the sample correlation coefficient

$$R^2 = r_{XY}^2.$$

Relationship between sample correlation coefficient and coefficient of determination

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since S_{yy} is exactly SST.

- Hence,

$$r_{XY}^2 = \hat{\beta}_1^2 \frac{S_{xx}}{SST} = \hat{\beta}_1 \frac{S_{xy}}{S_{xx}} \frac{S_{xx}}{SST} = \frac{\hat{\beta}_1 S_{xy}}{SST} = \frac{SSR}{SST} = R^2.$$

- Therefore, the coefficient of determination R^2 of the simple linear regression model is equal to the square of the sample correlation coefficient

$$R^2 = r_{XY}^2.$$

Note: R^2 here is a symbol, not the square of R .

Hypothesis testing for the correlation coefficient

- The hypothesis testing problem for the correlation coefficient ρ_{XY} of the simple linear regression model is formulated as follows:

$$\begin{cases} H_0 : \rho = 0 & \text{(no correlation between } X \text{ and } Y) \\ H_1 : \rho \neq 0 & \text{(there is correlation between } X \text{ and } Y) \end{cases}$$

with a specified significance level α .

Hypothesis testing for the correlation coefficient

Testing procedure

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3. Calculate the test statistic:

$$t_0 = r_{XY} \sqrt{\frac{n-2}{1-r_{XY}^2}}.$$

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6. Conclusion.

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- ▶ The assumptions are typically checked using the residuals, given by

$$e_i = y_i - \hat{y}_i, \quad i = 1, \dots, n,$$

with $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$.

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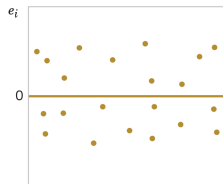
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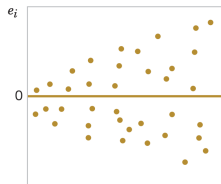
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- ▶ The residual plot also helps us identify the presence of **outliers**.
- ▶ To test the assumption of normal distribution (assumption 4), we often use the **Normal Q-Q Plot**.

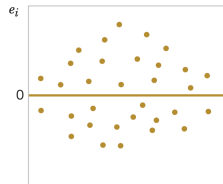
Residual analysis



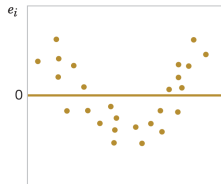
(a)



(b)



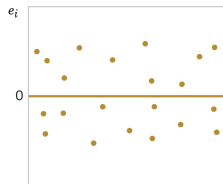
(c)



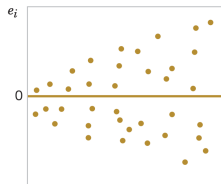
(d)

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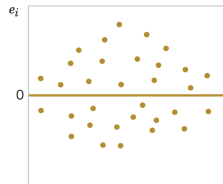
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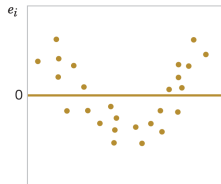
(a)



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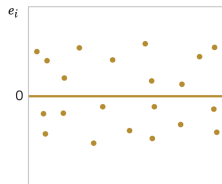
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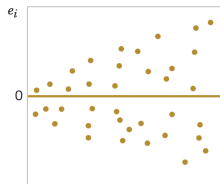
(d)

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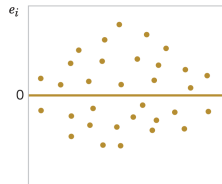
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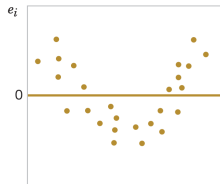
(a)



(b)



(c)

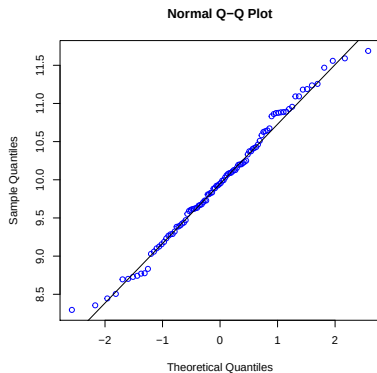


(d)

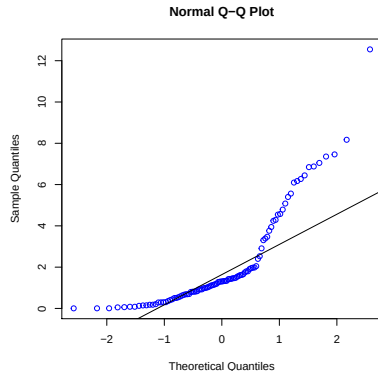
- ▶ (a): all assumptions of the model are satisfied.
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- (c): unequal variances.
- ▶ (d): non-linear relationship between X and Y .

Residual analysis

- ▶ Checking normal distribution using the **Normal Q-Q Plot**.



Data follows normal distribution



Data does not follow normal distribution

Exercises

Exercise 1

In a research paper on Concrete Research: "Surface characteristics of concrete: Intrinsic permeability" presents data on the compressive strength (X) and intrinsic permeability (Y) of different concrete mixes and various curing methods. The data are summarized as follows:

$$n = 14, \sum_{i=1}^n x_i = 43, \sum_{i=1}^n x_i^2 = 157.42, \sum_{i=1}^n y_i = 572, \\ \sum_{i=1}^n y_i^2 = 23530, \sum_{i=1}^n x_i y_i = 1697.80.$$

- (a) Determine the estimated regression line describing the linear relationship between compressive strength and intrinsic permeability of concrete.
- (b) Estimate the variance σ^2 of the residuals.
- (c) Using the estimated regression line, predict the intrinsic permeability of concrete when the compressive strength is $x_0 = 4.3$.
- (d) Calculate the coefficient of determination R^2 and provide comments on the relationship between X and Y .

Exercises

Exercise 2

Consider a sample of 10 pairs of values (x_i, y_i) given by the table

x_i	-1	0	2	-2	5	6	8	11	12	-3
y_i	-5	-4	2	-7	6	9	13	21	20	-9

- (a) Draw a scatter plot for the data, find the estimated regression line.
- (b) Find the estimate $\hat{\sigma}^2$ for the variance σ^2 of the random errors.
- (c) Calculate the coefficient of determination R^2 and the sample correlation coefficient r_{XY} .
- (d) Perform a hypothesis test for the coefficient β_1 .

Exercises

Exercise 3

A study investigates the effect of increasing dosage X (mg/kg) of a sleeping pill on sleep duration Y (hours). The experimental results are recorded as follows:

x_i	1	1	2	2	3	4	5	5
y_i	1	1.2	1.5	1.7	2	2.2	2.5	2.2

- (a) Find the regression equation of Y on X .
- (b) Find $\hat{\sigma}^2$ and the coefficient of determination R^2 .
- (c) If the dosage of the sleeping pill is $x_0 = 4$ (mg/kg), what is the predicted sleep duration?
- (d) The literature states that the regression equation of Y on X is $y = 0.29x + 0.93$. Does the observed result agree with the provided equation? $\alpha = 0.05$.